

Analysis PhD Exam, January 2026

Write solutions in a neat and logical fashion, giving complete reasons for all steps. Do six of eight.

1. Do two of three.

- (i) Give an example of a Lebesgue measurable set $E \subseteq [0, 1]$ that is closed, has positive measure, but contains no non-trivial interval or explain why no such example exists.
- (ii) Is the function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = \frac{1}{1+x^2}$ the Hardy–Littlewood maximal function of an $L^1(\mathbb{R})$ function f ?
- (iii) Does there exist a measure space $(X, \mathcal{M}, \mu)$ and a nested decreasing sequence $(E_n)_{n=1}^\infty$ of measurable sets such that

$$\lim_{n \rightarrow \infty} \mu(E_n) \neq \mu\left(\bigcap_{n=1}^\infty E_n\right)?$$

2. Suppose $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$. Show that if the following conditions hold, then $F : [0, 1] \rightarrow \mathbb{R}$ defined by

$$F(x) = \int_{[0,1]} f(x, y) d\lambda(y)$$

is continuous. Here λ denotes Lebesgue measure on $\mathbb{R}$.

- (a) For each $x \in [0, 1]$ the function $g_x(y) = f(x, y)$ is continuous;
- (b) For each $y \in [0, 1]$ the function $h_y(x) = f(x, y)$ is continuous; and
- (c) f is bounded.

3. Let $(X, \mathcal{M})$ denote a measurable space and let $P(X)$ denote the power set of X . Prove, if $\mathcal{M} \neq P(X)$ and $\{x\} \in \mathcal{M}$ for each $x \in X$, then there exists an $A \notin \mathcal{M} \otimes \mathcal{M}$ such that $[A]_x, [A]^y \in \mathcal{M}$ for all $x, y \in X$. Suggestion: First show $\delta : X \rightarrow X \times X$ defined by $\delta(x) = (x, x)$ is measurable from $(X, \mathcal{M})$ to $(X \times X, \mathcal{M} \otimes \mathcal{M})$.

4. Suppose $(X, \mathcal{M}, \mu)$ is a measure space and $f, g : X \rightarrow [0, \infty)$ are measurable. Show:

- (a) $\mu_f : \mathcal{M} \rightarrow [0, \infty]$ defined by $\mu_f(A) = \int \chi_A f d\mu = \int_A f d\mu$ is a measure; and
- (b) $\int g d\mu_f = \int gf d\mu$.

5. Do two of three. Short answer: you do not need to give a detailed proof, just a brief explanation.

- (a) If $a_1, \dots, a_n$ are nonnegative real numbers, must it be the case that

$$\left(\sum_{j=1}^n a_j\right)^6 \leq n^5 \sum_{j=1}^n a_j^6?$$

- (b) Explain why, given $x \in [0, 1]$, the mapping $E_x : C([0, 1]) \rightarrow \mathbb{C}$ defined by $E_x(f) = f(x)$ is a bounded linear functional. Is the dual of the Banach space $C([0, 1])$ separable?
- (c) Give an example where the Radon–Nikodym Theorem fails without the sigma-finite hypothesis.

6. Let $\mathbb{C}[x]$ denote the complex vector space of polynomials in x . Does there exist a norm $\|\cdot\|$ on $\mathbb{C}[x]$ such that $(\mathbb{C}[x], \|\cdot\|)$ is a Banach space. Provide a proof for your answer.

7. Suppose ϕ, ψ are bounded linear functionals on a Hilbert space V . Show, if $\|\phi + \psi\| = \|\phi\| + \|\psi\|$, then one of ϕ, ψ is a multiple of the other. Is the result still true if V is only assumed to be a Banach space?
8. Show if $f \in L^1(\mathbb{R})$ and $g \in L^\infty(\mathbb{R})$, then the convolution $f * g$ is continuous. Let $E \subseteq [0, 1]$ be a Lebesgue measurable set of positive measure and let

$$-E = \{-x : x \in E\}.$$

By considering the function $h = \chi_{-E} * \chi_E$ or otherwise, show

$$E - E = \{x - y : x, y \in E\}$$

contains an open interval.